



Lab 04 – Worksheet

Name: Faaz Shahab, Mohib Ul Mehdi, Basim Hussain	ID: fs09911, mm09730, mh09732	Section: T2
--	-------------------------------	-------------

Task 1

Provide appropriately commented codes.

```
addi x5, x0, 10 # n = 10
addi x6, x0, 1  # sum = 1

loop:
    mul x6, x6, x5
    addi x5, x5, -1
    bne x5, x0, loop

end:
    j end
```

**Add snip of relevant Registers / Memory locations where you think results are obtained. Make sure the irrelevant area of snip is cropped. Make sure the irrelevant area of snip is cropped.*

```
x06 (t1) = 3628800
```

Comments

In this task I implemented factorial using loop instead of recursion. Stack was not used because there were no nested function calls.

Task 2

Provide appropriately commented codes

```
li x5, 10
jal x6, faaz
li x10, 1
ecall

exit:
    j exit

faaz:
```



```
addi sp, sp, -8
sw x6, 4(sp)
sw x5, 0(sp)
addi x7, x5, -1
blt x7, x0, base_case
addi x5, x5, -1
jal x6, faaz
lw x28, 0(sp)
add x5, x5, x28
lw x6, 4(sp)
addi sp, sp, 8
jalr x0, 0(x6)

base_case:
li x5, 0
lw x6, 4(sp)
addi sp, sp, 8
jalr x0, 0(x6)
```

Add screenshot of your results

```
x05 (t0) = 55
x06 (t1) = 8
x07 (t2) = -1
```

Comments

This task helped me understand recursion and stack frames clearly.
Stack grows on every call and restores values correctly on return.



Task 3

Provide appropriately commented codes

```
addi sp, sp, -32 #32 bytes on stack for 8 integers
# [5,2,9,1,7,3,8,4]
li x8, 5
sw x8, 0(sp)
li x8, 2
sw x8, 4(sp)
li x8, 9
sw x8, 8(sp)
li x8, 1
sw x8, 12(sp)
li x8, 7
sw x8, 16(sp)
li x8, 3
sw x8, 20(sp)
li x8, 8
sw x8, 24(sp)
li x8, 4
sw x8, 28(sp)

mv x12, sp # base address
li x13, 8 # length
jal x1, bubble
addi sp, sp, 32 # restore stack
li x17, 10
ecall

# Bubble Sort
bubble:
    addi x6, x13, -1 # x6 = n-1 outer loop counter
    outer:
        beq x6, x0, exit # if passes finished, exit
        mv x7, x0 # j = 0

    inner:
        slli x28, x7, 2 # offset = j*4
        add x29, x12, x28
        lw x30, 0(x29) # a[j]
        lw x31, 4(x29) # a[j+1]
        ble x30, x31, noswap
        sw x31, 0(x29) # swap
        sw x30, 4(x29)

    noswap:
        addi x6, x6, -1
        jne x6, x0, outer
    exit:
        ret
```



```
noswap:
    addi x7, x7, 1
    blt x7, x6, inner # while j < n-1-i
    addi x6, x6, -1
    j outer

exit:
    j exit
```



Add screenshots of your results

0x7FFFFFFEC	04
0x7FFFFFFE8	08
0x7FFFFFFE4	03
0x7FFFFFFE0	07
0x7FFFFFFDC	01
0x7FFFFFFD8	09
0x7FFFFFFD4	02
0x7FFFFFFD0	05

Sorted:

0x7FFFFFFEC	09
0x7FFFFFFE8	08
0x7FFFFFFE4	07
0x7FFFFFFE0	05
0x7FFFFFFDC	04
0x7FFFFFFD8	03
0x7FFFFFFD4	02
0x7FFFFFFD0	01



Task 4

Provide appropriately commented codes

```
li x10, 2
li x11, 5
jal x1, power
j exit

power:
    addi sp, sp, -8
    sw x1, 0(sp)
    sw x10, 4(sp)
    beq x11, x0, base
    addi x11, x11, -1
    jal x1, power
    lw x5, 4(sp)
    lw x1, 0(sp)
    addi sp, sp, 8
    mul x10, x10, x5
    jalr x0, 0(x1)

base:
    li x10, 1
    lw x1, 0(sp)
    addi sp, sp, 8
    jalr x0, 0(x1)

exit:
    j exit
```



Add screenshots of your results

```
x00 (zero) = 0
x01 (ra)   = 0
x02 (sp)   = 2147483632
x03 (gp)   = 268435456
x04 (tp)   = 0
x05 (t0)   = 0
x06 (t1)   = 0
x07 (t2)   = 0
x08 (s0)   = 0
x09 (s1)   = 0
x10 (a0)   = 2
x11 (a1)   = 5
x10 (a0)   = 32
```



Assessment Rubric

Lab 04: Implementing Nested Procedures and Sorting

Task No.	LR 2 Code	LR 5 Results
Task 1	/10	/05
Task 2	/10	/05
Task 3	/15	/10
Task 4	/15	/10
Total Points: 100	/50	/30
CLO Mapped	CLO 2	

Affective Domain	Rubric	Points	CLO Mapped
AR7	Report Submission & Git Upload	/10 & /10	CLO2

CLO	Total Points	Points Obtained
2	100	
Total	100	

For description of different levels of the mapped rubrics, please refer to the Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics provided here.



Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory	Level 2: Developing	Level 3: Good	Level 4: Exemplary
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/simulation model/network model doesnot implement the requiredfunctionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and doesnot produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full commandon the basic tools of the software.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorlyconstructed with erroneousresults. Titles, captions, units are not mentioned. Data is presented in anobscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are notaccurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minorerrors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captionsand proper units are mentioned. Data presentation is systematic.
AR9	Report Content/Code Comments	No summary provided. The number/amount of tasks completed below the level of satisfaction and/or submitted late	Couldn't provide good summary of in-lab tasks. Some major tasks were completed but not explained well. Submission on time. Some major plots andfigures provided	Good summary of In-lab tasks. All major tasks completed except few minor ones. The work is supported by some decent explanations, Submission on time, All necessary plots, and figures provided	Outstanding Summary of In-Lab tasks. All task completed and explained well, submitted on time, good presentation of plotsand figure with proper label, titles and captions